

FPGA Development for Radar, Radio-Astronomy and Communications

THE
RADAR
MASTERS COURSE



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Private Bag, Rondebosch, 7701, South Africa
<http://www.rmsg.uct.ac.za>



Presented by Dr John-Philip Taylor
Convened by Dr Stephen Paine

Day 2 – 4 August 2026

Outline

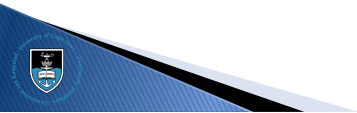
Finite State Machines

Simulation

JTAG

Timing Constraints

Advanced Timing Constraints



Outline

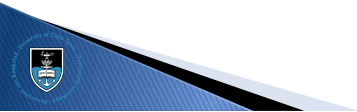
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Finite State Machines

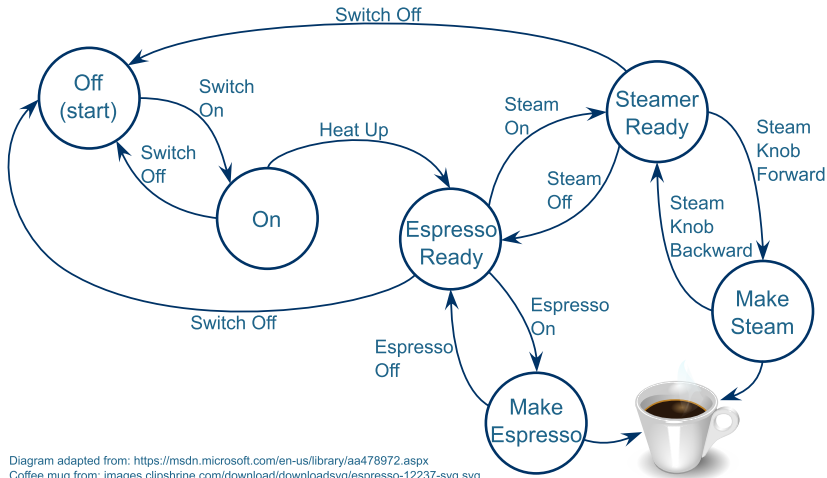


Diagram adapted from: <https://msdn.microsoft.com/en-us/library/aa478972.aspx>
Coffee mug from: images.clipshrine.com/download/downloadsvg/espresso-12237-svg.svg

Named States

► Example using Gray code:

```
reg [2:0] State;
```

```
localparam Off          = 3'b000;
```

```
localparam On           = 3'b001;
```

```
localparam EspressoReady = 3'b011;
```

```
localparam SteamerReady  = 3'b010;
```

```
localparam MakeEspresso  = 3'b110;
```

```
localparam MakeSteam     = 3'b111;
```



Named States

- Example using one-hot encoding:

```
reg [5:0] State;
```

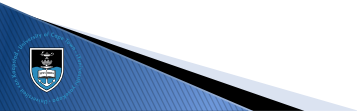
```
localparam Off           = 6'b000001;  
localparam On            = 6'b000010;  
localparam EspressoReady = 6'b000100;  
localparam SteamerReady  = 6'b001000;  
localparam MakeEspresso  = 6'b010000;  
localparam MakeSteam     = 6'b100000;
```



Named States

- Or let the compiler select the encoding:

```
typedef enum{ // SystemVerilog only
    Off,
    On,
    EspressoReady,
    SteamerReady,
    MakeEspresso,
    MakeSteam
} STATE;
STATE State;
```



FSM Template

```
always @(posedge ipClk) begin
    if(Reset) begin
        State <= Off;

    end else begin
        case(State)
            Off:          begin ... end
            On:           begin ... end
            EspressoReady: begin ... end
            SteamerReady: begin ... end
            MakeEspresso: begin ... end
            MakeSteam:    begin ... end
            default;;
        endcase
    end
end
```

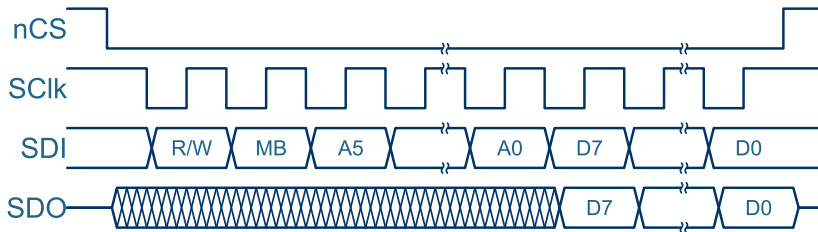


FSM Example

```
...  
case (State)  
  Off: begin  
    if (Switch_On) State <= On;  
  end  
  
  On: begin  
    if (Switch_Off) State <= Off;  
    else if (Heat_Up    ) State <= EspressoReady;  
  end  
  
  EspressoReady: begin  
    if (Switch_Off ) State <= Off;  
    else if (Steam_On    ) State <= SteamerReady;  
    else if (Espresso_On) State <= MakeEspresso;  
  end  
  
  ...
```



SPI Interface



- ▶ For the ADXL345 Digital Accelerometer:
 - ▶ Maximum SClk frequency is 5 MHz (200 ns period)
 - ▶ SDI setup and hold is 5 ns (sampled on rising edge)
 - ▶ SClk falling edge to SDO delay is 40 ns

The Abstraction

```
module ADXL345 #(
    parameter Clock_kHz,
    parameter Baud_kHz = 5_000
) (
    input ipClk, ipReset,

    // 2's Complement Output
    output reg [15:0] opX,
    output reg [15:0] opY,
    output reg [15:0] opZ,

    // Physical device interface
    output reg opnCS, opSClk, opSDI,
    input      ipSDO
);
localparam ClockDiv = Clock_kHz / Baud_kHz / 2;
```



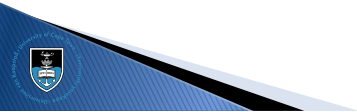
General Structure

```
reg      Reset;  
reg [3:0]ClockCount  = 0;  
wire     ClockEnable = (ClockCount == ClockDiv);  
  
always @(posedge ipClk) begin  
    Reset <= ipReset;  
  
    if(ClockEnable) ClockCount <= 4'd1;  
    else             ClockCount <= ClockCount + 1'b1;  
  
    if(Reset) begin  
        // Reset the machine here  
  
    end else if(ClockEnable) begin  
        // State machine goes here  
    end  
end
```



State Machine

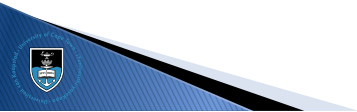
```
reg [ 4:0]Count;  
reg [15:0]Data; // (R/W, MB, Address, Byte) or (2 Bytes)  
  
typedef enum {  
    Setup,  
    ReadX, ReadY, ReadZ,  
    Transaction  
} STATE;  
  
STATE State;  
STATE RetState; // Used for function calls
```



State Machine

```
if(Reset) begin
    opnCS    <= 1'b1;
    opSClk   <= 1'b1;
    opSDI    <= 1'b1;
    State    <= Setup;

end else if(ClockEnable) begin
    case(State)
        Setup: begin
            // SPI 4-wire; Full-res; Right-justify; 4g Range
            Data      <= {2'b00, 6'h31, 8'b0000_1001};
            Count     <= 5'd16;
            State     <= Transaction;
            RetState  <= ReadX;
        end
    end
```



State Machine

ReadX: **begin**

opZ <= {Data[7:0], Data[15:8]};

Data <= {2'b11, 6'h32, 8'd0};

Count <= 5'd24;

State <= Transaction;

RetState <= ReadY;

end

ReadY: **begin**

opX <= {Data[7:0], Data[15:8]};

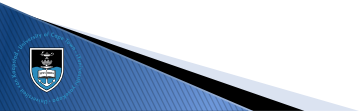
Data <= {2'b11, 6'h34, 8'd0};

Count <= 5'd24;

State <= Transaction;

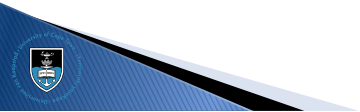
RetState <= ReadZ;

end



State Machine

```
ReadZ: begin  
    opY      <= {Data[7:0], Data[15:8]};  
    Data     <= {2'b11, 6'h36, 8'd0};  
    Count    <= 5'd24;  
    State    <= Transaction;  
    RetState <= ReadX;  
end
```



Reading Data

```
Transaction: begin
    if(opnCS) begin
        opnCS <= 1'b0;

    end else begin
        if(opSClk) begin
            if(Count == 0) begin
                opnCS <= 1'b1;
                State <= RetState;
            end else begin
                opSClk <= 1'b0;
            end
            Count <= Count - 1'b1;
            {opSDI, Data} <= {Data, ipSDO};
        end else begin
            opSClk <= 1'b1;
        end end end
```



Outline

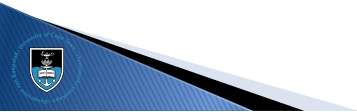
Finite State Machines

Simulation

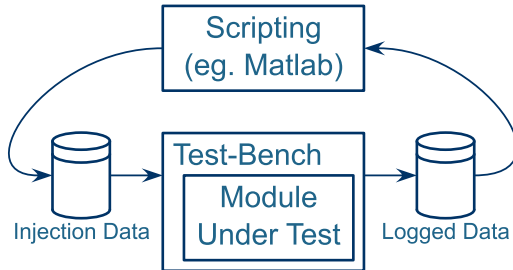
JTAG

Timing Constraints

Advanced Timing Constraints

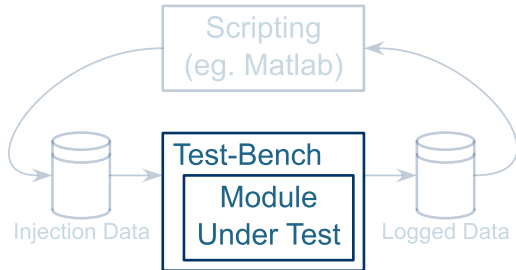


Simulation Basics

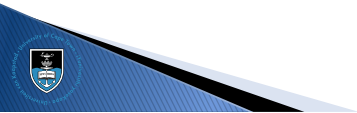


Today we only focus on the test-bench and simulation tool

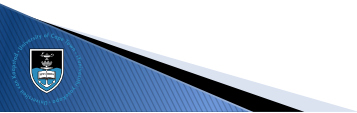
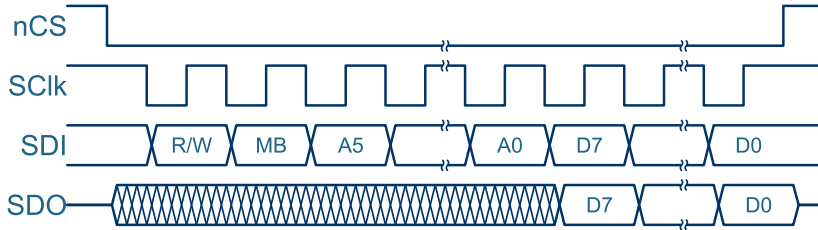
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ADXL345 SPI Interface

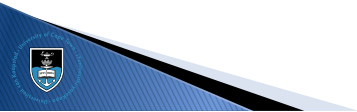


Test Bench Basics

```
`timescale 1ns/1ps
module ADXL345_TB;

// Clock
reg ipClk = 0;
always #10 ipClk <= ~ipClk;

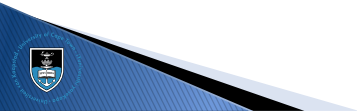
// Reset
reg ipReset = 1;
initial #50 ipReset <= 0;
```



Test Bench Basics

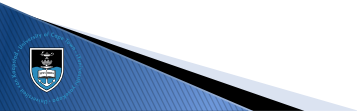
```
// DUT
wire [15:0]opX, opY, opZ;
wire opnCS, opSClk, opSDI;
reg ipSDO = 0;

ADXL345 #(50_000) Accelerometer( // Set parameter for 50 MHz clock
    ipClk, ipReset,
    opX, opY, opZ,
    opnCS, opSClk, opSDI, ipSDO
);
```



Test Bench Basics

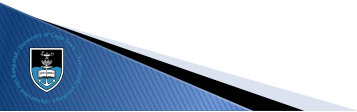
```
reg [ 7:0]DataIn;  
reg [15:0]DataOut = 0;  
  
integer n;  
always begin  
    @(negedge opnCS);  
  
    // Instruction word  
    for(n = 7; n >= 0; n--) begin  
        @(negedge opSClk);  
        DataIn[n] <= opSDI;  
    end
```



Test Bench Basics

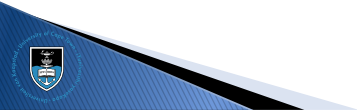
```
// The first data word
for(n = 7; n >= 0; n--) begin
    @(negedge opSClk); #40 // Output delay;
    ipSDO <= DataOut[n];
end

// The optional second data word
if(DataIn[6]) begin // More bits
    for(n = 15; n >= 8; n--) begin
        @(negedge opSClk); #40 // Output delay;
        ipSDO <= DataOut[n];
    end
end
```



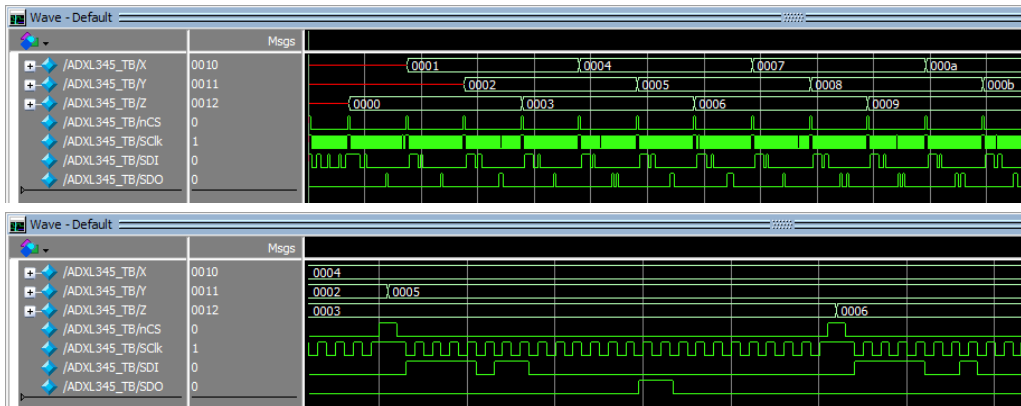
Test Bench Basics

```
@(posedge opnCS);  
  
    DataOut <= DataOut + 1;  
end  
  
endmodule
```



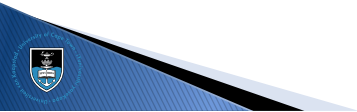
Questa

The resulting Questa waveforms:



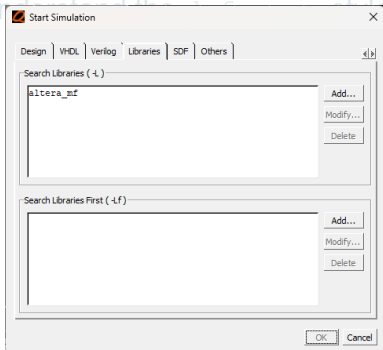
Simulation

- ▶ Questa can simulate IP modules
- ▶ Add the `altera_mf` library in the “Start Simulation” dialogue box
- ▶ Also remember to compile the IP block
- ▶ Questa cannot understand the `defparam` style of parameters.



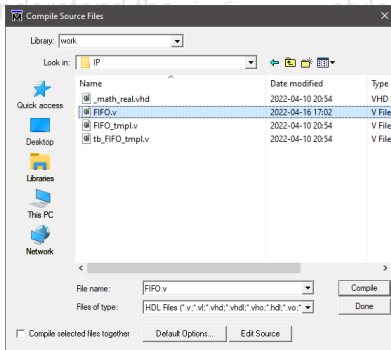
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Simulation

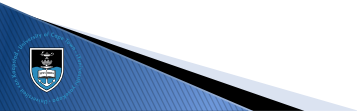
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Simulation

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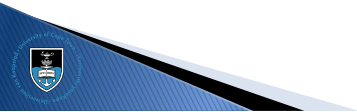
```
Mod Inst(  
    // Port assignments  
);  
defparam Inst.Param1 = 8'h12;  
defparam Inst.Param2 = 8'h23;
```



Simulation

- ▶ Questa can simulate IP modules
- ▶ Add the `altera_mf` library in the “Start Simulation” dialogue box
- ▶ Also remember to compile the IP block
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```
Mod # (  
    .Param1(8'h12),  
    .Param2(8'h23)  
) Inst (  
    // Port assignments  
);
```



Outline

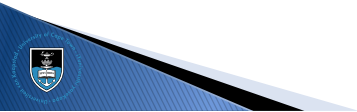
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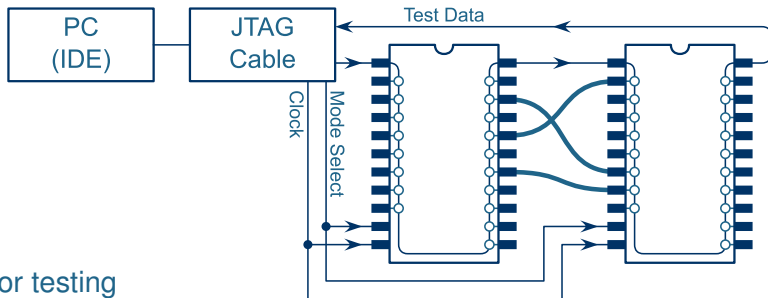
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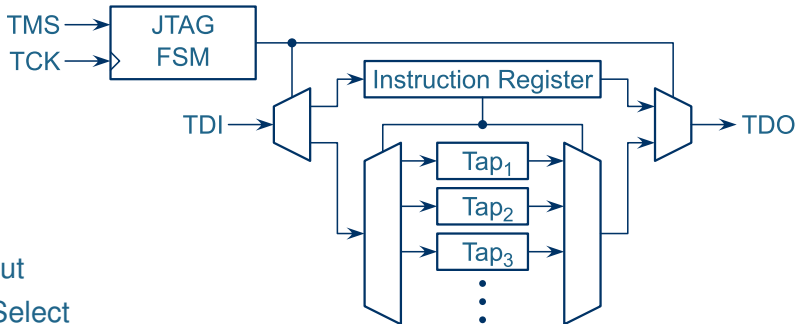


JTAG – Joint Test Action Group

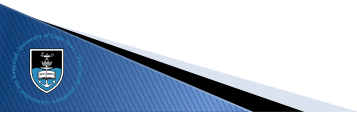


- ▶ Originally intended for testing soldering quality on populated PCBs
- ▶ Connects to the PC over USB / Ethernet / etc.
- ▶ Connected devices form a long chain of shift-registers

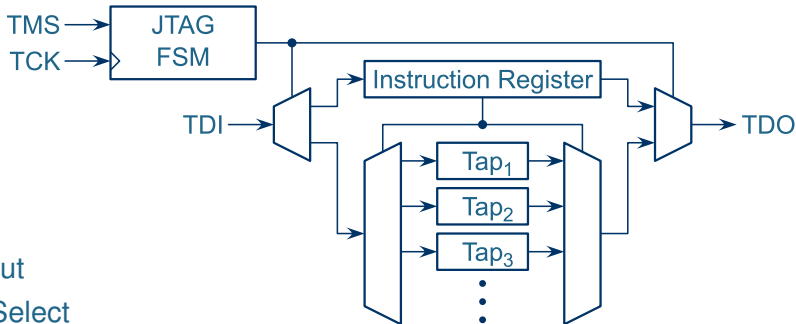
JTAG – Joint Test Action Group



- ▶ TCK: Test Clock
- ▶ TDI: Test Data In
- ▶ TDO: Test Data Out
- ▶ TMS: Test Mode Select
- ▶ Taps could include: Device pins; Internal flash; Status registers; Debug registers; Virtual JTAG interface; etc.



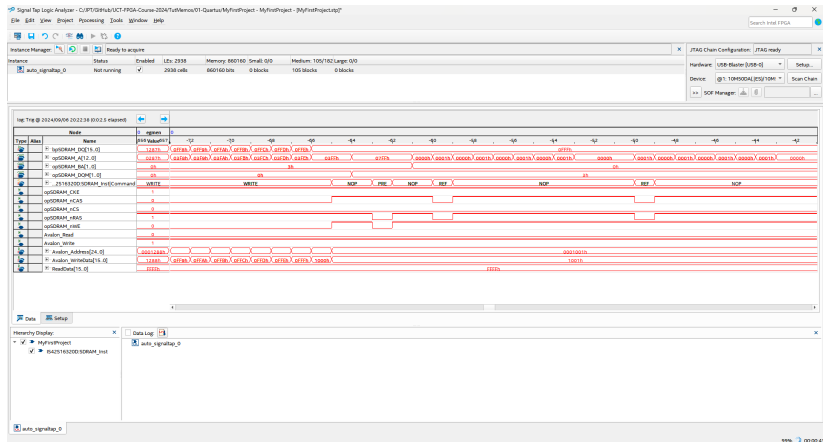
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JTAG Debugging

FPGA IDEs includes powerful JTAG-based debugging tools.



JTAG Debugging

FPGA IDEs includes powerful JTAG-based debugging tools.

- ▶ Sources and Probes

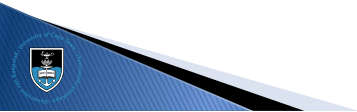
<https://www.youtube.com/watch?v=Mftgi318Nrc>

- ▶ In-system memory editor

https://www.youtube.com/watch?v=_VcVtFvJnuY

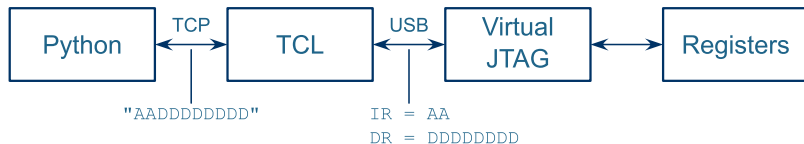
- ▶ Signal-tap logic analyser

<https://www.youtube.com/watch?v=vhkzxCEXuaA>

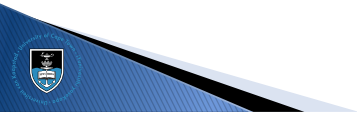


JTAG Debugging

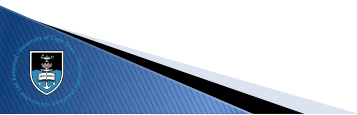
You can do some really interesting things using TCL scripting and the JTAG interface.



Idle Logic Labs: Talking to the DE0-Nano using the Virtual JTAG interface



Coffee Break...



Outline

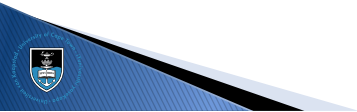
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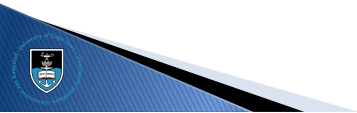
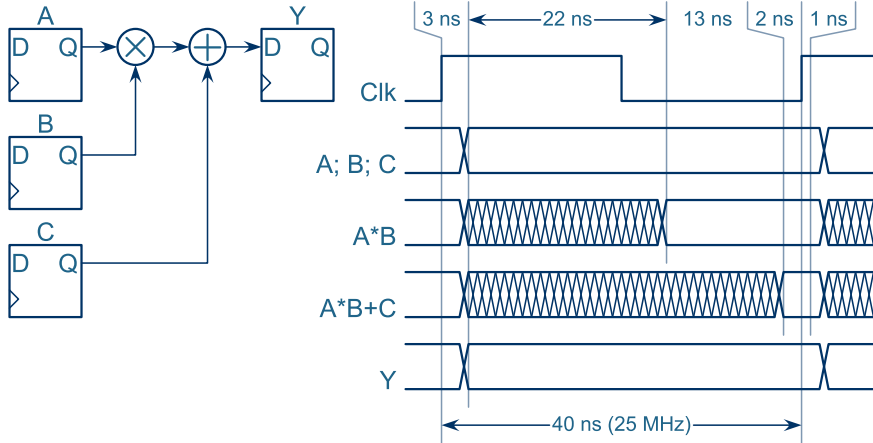
JTAG

Timing Constraints

Advanced Timing Constraints

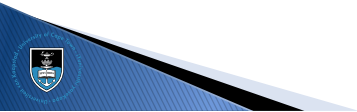


Internal Timing



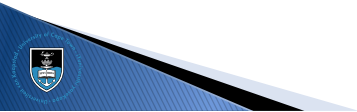
Synopsis Design Constraints

- ▶ De facto industry standard
- ▶ TCL based, so one can use TCL scripting within the SDC file
- ▶ Only specify what the compiler does not already know:
 - ▶ Clock frequencies
 - ▶ Asynchronous paths
 - ▶ PCB trace delays
 - ▶ External device parameters
 - ▶ Multi-cycle paths
 - ▶ etc.



Synopsis Design Constraints

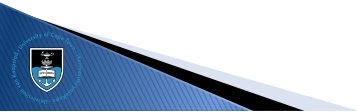
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Asynchronous Pins

- ▶ Pins such as LEDs, buttons, RS-232 signals, etc. does not belong to a clock domain
- ▶ Quartus must not try to meet timing on these:

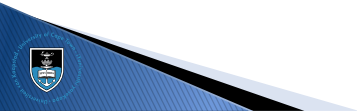
```
set_false_path -from * -to [get_ports opLED*]  
set_false_path -to * -from [get_ports ipSwitch*]  
set_false_path -to * -from [get_ports ipButton*]
```



Asynchronous Pins

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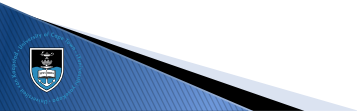
Clock Specification

```
create_clock -period 100 [get_ports ADC_CLK_10]
create_clock -period 20 [get_ports MAX10_CLK1_50]
create_clock -period 20 [get_ports MAX10_CLK2_50]

derive_pll_clocks -create_base_clocks -use_net_name
derive_clock_uncertainty

create_generated_clock \
  -source [get_pins { SDRAM_PLL_Inst|*|clk[1] } ] \
  -name opClk_SDRAM [get_ports opClk_SDRAM]
```

To get the node names, use the “Clocks” report of the Timing Analyzer and copy the name, or use the Timing Analyzer GUI to search for it.



Clock Specification

```
create_clock -period 100 [get_ports ADC_CLK_10]
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Clock Report

Quantus Prime Lite Edition - C:\PT\Projects\161608 - FPGA Master Course\Quantus\New Project\MyFirstProject - MyFirstProject

File Edit View Project Assignments Processing Tools Window Help

MyFirstProject

Project Navigator

Entity Instance

- MAX10_10M50QAF48C7G
 - MyFirstProject
 - MySys:MySys_inst
 - altsource_probeSourcesAndProbes_inst
 - slk_hubaudio_hub

Table of Contents

- Flow Summary
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- TimeQuest Timing Analyzer
 - Summary
 - Parallel Compilation
 - SDC File List
 - Clocks
 - Slow 1200mV BSC Model
 - Slow 1200mV OC Model
 - Fast 1200mV OC Model
 - Multicore Timing Analysis Summary
 - Advanced I/O Timing
 - Clock Transfers
 - Report TCCS
 - Report RSKM
 - Unconstrained Paths
 - Messages
 - EDA Netlist Writer
 - Flow Messages
 - Flow Suppressed Messages

Compilation Report - MyFirstProject

Clocks

	Clock Name	Type	Period	Frequency	Rise	Fall	Duty Cycle	Divide by	Multiply by	Phase
1	ADC_CLK_10	Base	100.000	10.0 MHz	0.000	50.000				
2	alters_reserved_tick	Base	100.000	10.0 MHz	0.000	50.000				
3	DRAM_CLK	Generated	10.000	100.0 MHz	7.500	12.500		1	1	
4	MAX10_CLK1_50	Base	20.000	50.0 MHz	0.000	10.000				
5	MAX10_CLK2_50	Base	20.000	50.0 MHz	0.000	10.000				
6	MySys:MySys_inst...[wive_p87_clk[0]	Generated	10.000	100.0 MHz	0.000	50.00	50.00	1	2	
7	MySys:MySys_inst...[wive_p87_clk[1]	Generated	10.000	100.0 MHz	7.500	12.500	50.00	1	2	-90.0

Tasks

Task	Time
Compile Design	00:01:35
Analysis & Synthesis	00:00:56
Fitter (Place & Route)	00:00:21
Assembler (Generate programming files)	00:00:08
TimeQuest Timing Analysis	00:00:06
EDA Netlist Writer	00:00:04
Edit Settings	
Program Device (Open Programmer)	

IP Catalog

Installed IP

- Project Directory
- System
- Basic Functions
- DSP
- Interface Protocols
- Memory Interfaces and Controllers
- Processors and Peripherals
- University Program
- Search for Partner IP

Find... Find Next

Messages

System Processing (219)

100% 00:01:35

18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NIM_PARALLEL_PROCESSORS in your QSF to an appropriate value for best performance.

10995 Generated the EDA functional simulation netlist because it is the only supported netlist type for this device.

204019 Generated file MyFirstProject.svo in folder "C:\PT\Projects\161608 - FPGA Master Course\Quantus\New Project\simulation\modelsim\" for EDA simulation tool

Quantus Prime EDA Netlist writer was successful. 0 errors, 2 warnings

293000 Quantus Prime Full compilation was successful. 0 errors, 34 warnings



Clock Groups

- ▶ Unless specified otherwise, Quartus assumes that all clocks are related and in the same clock domain
- ▶ When clocks are unrelated, all paths between them must be marked as “false paths”
- ▶ Do this with clock groups:

```
set_clock_groups -logically_exclusive \  
  -group [get_clocks ADC_CLK_10]      \  
  -group [get_clocks MAX10_CLK1_50]   \  
  -group [get_clocks MAX10_CLK2_50]   \  
  -group [get_clocks {opClk_SDRAM *altpll_component*}]
```



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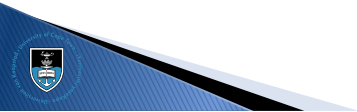
```
set_clock_groups -logically_exclusive \  
  -group [get_clocks ADC_CLK_10]      \  
  -group [get_clocks MAX10_CLK1_50]   \  
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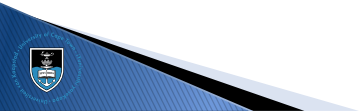
```
set_clock_groups -logically_exclusive \  
-group [get_clocks ADC_CLK_10]      \  
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```



JTAG Timing Specification

- ▶ Standard across all Altera FPGAs
- ▶ Copied from Altera examples:

```
# Don't specify the altera_reserved_tck frequency,  
# the compiler knows what it is, but put it in its own group  
set_clock_groups -exclusive -group [get_clocks {altera_reserved_tck}]  
  
set_input_delay -clock altera_reserved_tck \  
    -clock_fall 3 [get_ports {altera_reserved_tdi}]  
set_input_delay -clock altera_reserved_tck \  
    -clock_fall 3 [get_ports {altera_reserved_tms}]  
set_output_delay -clock altera_reserved_tck \  
    -clock_fall 3 [get_ports {altera_reserved_tdo}]
```



Outline

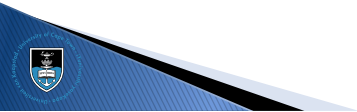
Finite State Machines

Simulation

JTAG

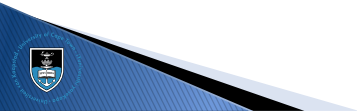
Timing Constraints

Advanced Timing Constraints



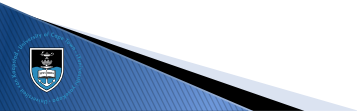
External Timing Requirements

- ▶ Synopsis Design Constraints start with the ideal case:
 - ▶ There are no PCB trace delays
 - ▶ The external setup requirement is zero
 - ▶ The external hold requirement is zero
 - ▶ The external propagation delay is zero
- ▶ Increase the maximum output delay for external setup timing
- ▶ Decrease the minimum output delay for external hold timing
- ▶ Specify the minimum and maximum input delays according to the external propagation delay parameters
- ▶ Worsen the situation with PCB trace delays and uncertainties (clock jitter, manufacturing tolerances, etc.)

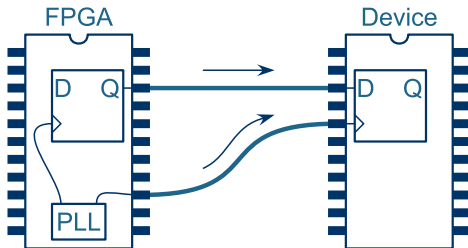


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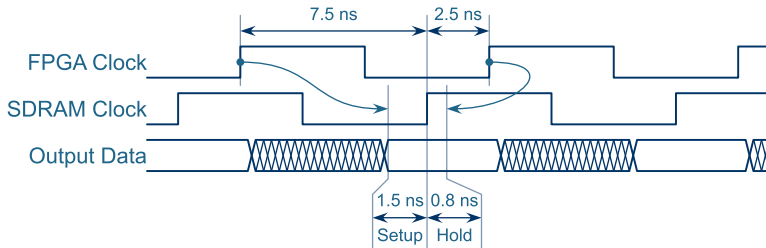


External Timing – Output

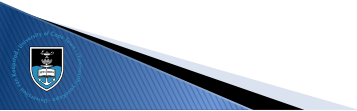


- ▶ FPGA internal delays and PCB trace delays cancel
- ▶ The important parameters are:
 - ▶ Setup and hold times of the external device
 - ▶ Clock jitter and other uncertainties
- ▶ Shift the external clock to ease the hold margin

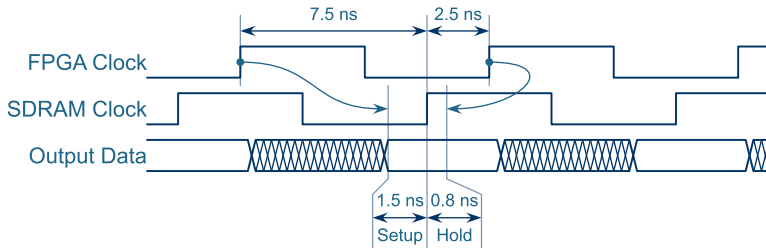
External Timing – Output



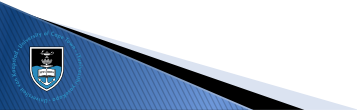
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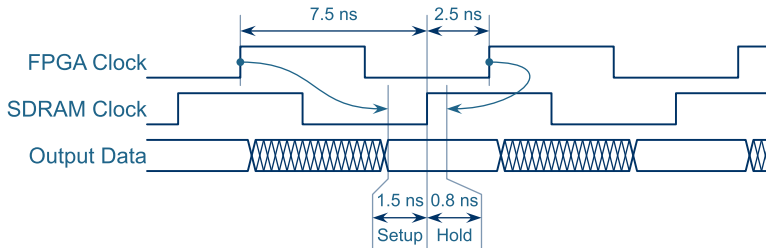
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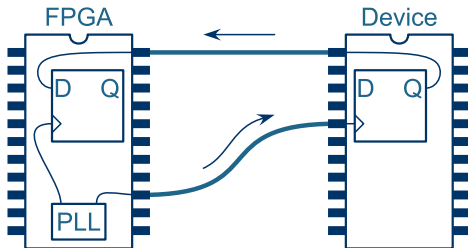
External Timing – Output



Suppose 100 ps uncertainty

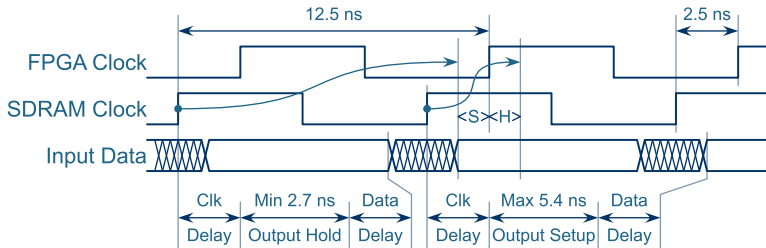
```
set_output_delay -max -clock opClk_SDRAM 1.6 [get_ports opSDRAM*]
set_output_delay -min -clock opClk_SDRAM -0.9 [get_ports opSDRAM*]
set_output_delay -max -clock opClk_SDRAM 1.6 [get_ports bpSDRAM*]
set_output_delay -min -clock opClk_SDRAM -0.9 [get_ports bpSDRAM*]
```

External Timing – Input

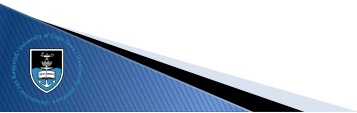


- ▶ Large FPGA internal delays and PCB trace delays
- ▶ Shift the external clock to ease the setup margin
- ▶ Multi-cycle requirement on the setup path (otherwise the compiler uses the 2.5 ns path) – the minimum propagation delay of 2.7 ns makes the 2.5 ns clock shift safe

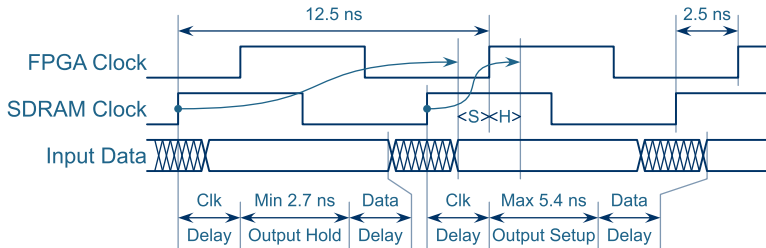
External Timing – Input



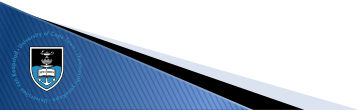
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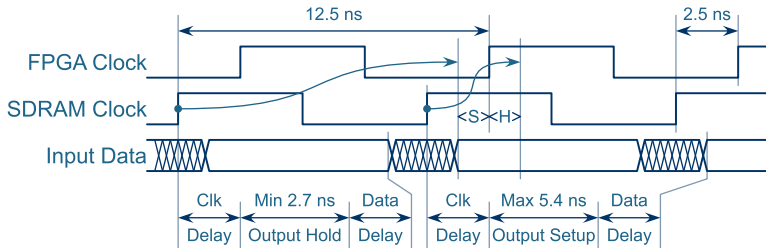
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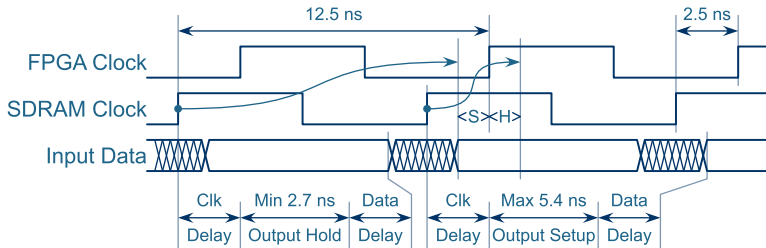


External Timing – Input



```
set_multicycle_path \
  -from [get_clocks opClk_SDRAM] \
  -to   [get_clocks SDRAM_PLL:SDRAM_PLL_Inst|*|wire_pll1_clk[0] ] \
  -setup 2
```


External Timing – Input

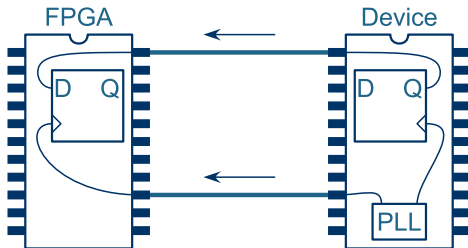


Suppose 100 ps uncertainty and 200 ps PCB delay (each way)

```
set_input_delay -max -clock opClk_SDRAM 5.9 [get_ports bpSDRAM*]
```

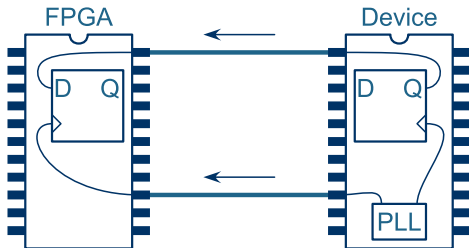
```
set_input_delay -min -clock opClk_SDRAM 3.0 [get_ports bpSDRAM*]
```

External Timing – DDR ADC

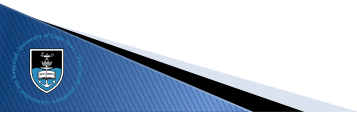


- ▶ The clock is sourced by the external device
- ▶ The PCB trace delays cancel
- ▶ The data is centre-aligned
- ▶ Use the dedicated serial LVDS receivers (on most FPGAs)

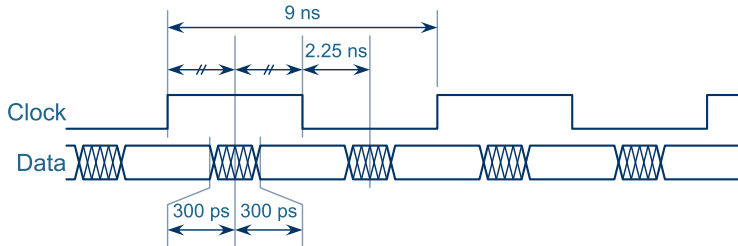
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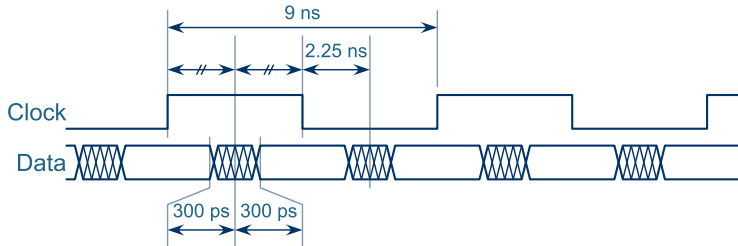


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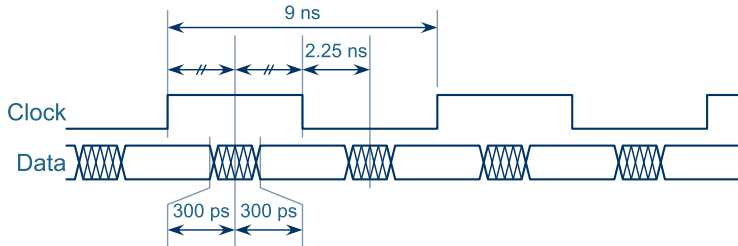
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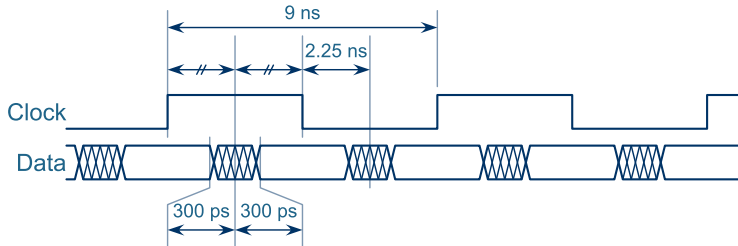
External Timing – DDR ADC



Suppose 100 ps uncertainty

```
create_clock      -name  ADC_DCO -period 9 [get_ports ipADC_DCLK_P]
set_input_delay  -clock  ADC_DCO -min 1.85 [get_ports ipADC_CH*]
set_input_delay  -clock  ADC_DCO -max 2.65 [get_ports ipADC_CH*]
```

External Timing – DDR ADC

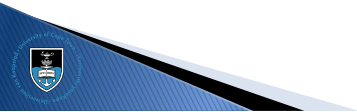


```
set_input_delay -clock ADC_DCO -clock_fall \  
-min 1.85 [get_ports ipADC_CH*] -add_delay  
set_input_delay -clock ADC_DCO -clock_fall \  
-max 2.65 [get_ports ipADC_CH*] -add_delay
```

Timing-related Pin Attributes

- ▶ Use the correct I/O standard
- ▶ Set the correct output current
- ▶ Set the pin capacitance

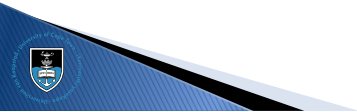
```
set_instance_assignment -name IO_STANDARD "3.3-V LVTTL" -to opClk_SDRAM  
set_instance_assignment -name IO_STANDARD "3.3-V LVTTL" -to opSDRAM*  
set_instance_assignment -name IO_STANDARD "3.3-V LVTTL" -to bpSDRAM*
```



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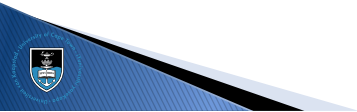
```
set_instance_assignment -name CURRENT_STRENGTH_NEW 8MA -to opClk_SDRAM  
set_instance_assignment -name CURRENT_STRENGTH_NEW 8MA -to opSDRAM*  
set_instance_assignment -name CURRENT_STRENGTH_NEW 8MA -to bpSDRAM*
```



Timing-related Pin Attributes

- ▶ Use the correct I/O standard
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- ▶ Set the pin capacitance

```
set_instance_assignment -name BOARD_MODEL_NEAR_C 3.5P -to opClk_SDRAM  
set_instance_assignment -name BOARD_MODEL_NEAR_C 3.8P -to opSDRAM*  
set_instance_assignment -name BOARD_MODEL_NEAR_C 6.0P -to bpSDRAM*
```



Select References



Stephen Brown and Zvonko Vranesic
Fundamentals of Digital Logic with Verilog Design, 2nd Edition
ISBN 978-0-07-721164-6



Merrill L Skolnik
Introduction to RADAR Systems
ISBN 978-0-07-288138-7



Mark A. Richards and James A. Scheer
Principles of Modern Radar: Basic Principles
ISBN 978-1-89-112152-4



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<http://www.asic-world.com/>



Jean P. Nicolle
FPGA 4 Fun
<http://www.fpga4fun.com/>

FPGA Development for Radar, Radio-Astronomy and Communications

THE
RADAR
MASTERS COURSE



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Presented by Dr John-Philip Taylor
Convened by Dr Stephen Paine

Day 2 – 4 August 2026